FRONT CHECK VALVE (Continued)

OPERATION

The front check valve provides more than one function in this application. It prevents washer fluid from draining out of the front washer supply hoses back to the washer reservoir. This drain-back would result in a lengthy delay from when the front washer switch is actuated until washer fluid was dispensed through the front washer nozzle, because the front washer pump would have to refill the front washer plumbing from the reservoir to the nozzle. Such a drain-back condition could also result in water, dirt, or other outside contaminants being siphoned into the washer system through the washer nozzle orifice. This water could subsequently freeze and plug the nozzle, while other contaminants could interfere with proper nozzle operation and cause improper nozzle spray patterns. In addition, the front check valve prevents washer fluid from siphoning out through the front washer nozzle after the front washer system is turned Off.

When the front washer pump pressurizes and pumps washer fluid from the reservoir through the front washer plumbing, the fluid pressure unseats a diaphragm from over a sump well within the valve by overriding the spring pressure applied to it by a piston (Fig. 4). With the diaphragm unseated, washer fluid is allowed to flow toward the front washer nozzle. When the front washer pump stops operating, the spring pressure on the piston seats the diaphragm over the sump well in the valve and fluid flow in either direction within the front washer plumbing is prevented. The front check valve cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

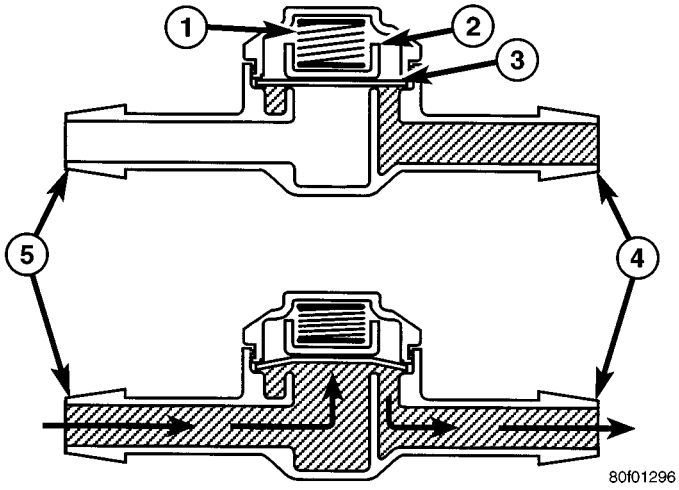

Fig. 4 Front Check Valve

1 - SPRING
2 - PISTON
3 - DIAPHRAGM
4 - TO WASHER NOZZLE
5 - FROM WASHER PUMP

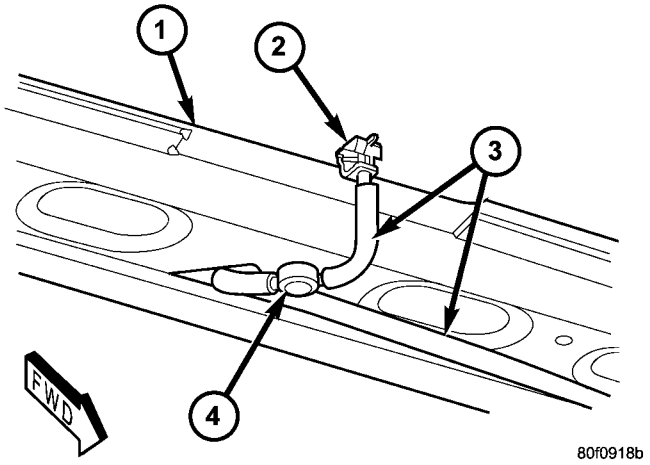

Fig. 5 Front Check Valve Remove/Install

1 - HOOD REAR REINFORCEMENT
2 - FRONT WASHER NOZZLE
3 - WASHER HOSE
4 - FRONT CHECK VALVE

REMOVAL

(1) Unlatch, open and support the hood.

(2) Locate the front check valve in the washer supply hose for the front washer nozzle. The check valve is installed in the washer supply hose about 5 centimeters (2 inches) from the front washer nozzle nipple near the rear hood panel reinforcement (Fig. 5).

(3) Disconnect the washer supply hose for the front washer nozzle from the barbed outlet nipple on the front washer system check valve. A small quantity of washer fluid may drain from the disconnected hose.

(4) Disconnect the washer supply hose for the reservoir from the barbed inlet nipple of the front check valve. Either install a temporary plug in the washer supply hose for the reservoir or secure the loose end of this hose at a point higher than the washer reservoir to prevent the contents of the washer reservoir from draining through this hose.

(5) Remove the front check valve from the engine compartment.